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Interface circuits for a six-phase variable reluctance energy harvester: a comparative study of rectification and power conditioning topologies

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*To my parents
and friends*

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Abstract

In recent years, more and more wireless sensor networks (WSN) and IoT devices are being used, and this has made their power supply a critical issue. Battery-based solutions have significant replacement and maintenance costs, especially when there are many sensor nodes or when they are hard to reach. Energy harvesting is the conversion of ambient energy, such as mechanical vibrations, heat, light, or RF signals, into usable electrical energy. It is proposed as a solution to this problem, since it allows the realization of autonomous nodes that need almost no maintenance.

This thesis starts with an overview of the main transduction technologies used for harvesting from mechanical energy. These include piezoelectric, electromagnetic, electrostatic and triboelectric technologies, and their advantages and limitations are discussed. The thesis then focuses on the technology based on variable reluctance, called Variable Reluctance Energy Harvester (VREH). This technology works well for low-speed rotational applications, because both the magnet and the coil stay fixed, and the change in linked flux is caused only by the passage of a ferromagnetic toothed wheel. The state of the art of research on this technology is then reviewed, starting from the first prototypes built for railway monitoring, moving through single-pole topologies, and arriving at the multi-phase system with six units (MP-VREH) used as reference in this work. This system is designed to be integrated into large commercial vehicles, such as buses, at rotation speeds between 100 and 400 rpm (about 20-80 km/h).

Starting from Faraday's law, a complete electrical model of the VREH is then built. This model gives the relations that link the extracted power to the geometric and electrical parameters of the transducer, such as the number of turns, the coil resistance, the flux difference between the aligned and unaligned position, and the number of teeth of the wheel. From this model, the maximum power that can theoretically be transferred to the load is found, and the conditions for impedance matching are discussed as a function of the rotation speed.

The central part of the thesis focuses on the design of the power conditioning circuits for the six-phase system considered. Several rectification topologies are proposed and compared. They are first evaluated through circuit simulation, and then through an experimental test bench, which includes the design of a dedicated PCB and the calibration of the current sensors needed to accurately measure the power extracted from the system. Finally, the most suitable power management integrated circuit (PMIC) is chosen to regulate and store the harvested energy. This completes the conversion chain, from rotational mechanical energy to a stable power supply for the final load.

Sommario

Negli ultimi anni la crescente diffusione di reti di sensori wireless (WSN) e di dispositivi IoT ha reso critico il problema della loro alimentazione. Le soluzioni a batteria comportano infatti costi di sostituzione e manutenzione non trascurabili, specialmente quando i nodi sensore sono numerosi o di difficile accesso. L'*energy harvesting*, ossia la conversione di energia ambientale (vibrazioni meccaniche, calore, luce, segnali RF) in energia elettrica utilizzabile, si propone come soluzione a questo problema, permettendo la realizzazione di nodi autonomi e a manutenzione pressoché nulla.

Questo lavoro di tesi si apre con una panoramica delle principali tecnologie di trasduzione, piezoelettrica, elettromagnetica, elettrostatica e triboelettrica, per l'harvesting da energia meccanica, evidenziandone vantaggi e limiti, per poi concentrarsi sulla tecnologia basata sulla riluttanza variabile (*Variable Reluctance Energy Harvester*, VREH). Questa tecnologia è particolarmente adatta alle applicazioni rotazionali a bassa velocità, poiché magneti e bobina restano entrambi fissi e la variazione di flusso concatenato è generata unicamente dal passaggio di una ruota dentata ferromagnetica. Viene quindi ripercorso lo stato dell'arte della ricerca su questa tecnologia, dai primi prototipi per il monitoraggio ferroviario, alle topologie a singolo polo, fino al sistema multifase (MP-VREH) a sei unità preso come riferimento in questo lavoro, pensato per l'integrazione in veicoli commerciali di grandi dimensioni, quali autobus, a velocità di rotazione comprese tra 100 e 400 rpm (circa 20-80 km/h).

A partire dalla legge di Faraday viene poi sviluppato un modello elettrico completo del VREH, con la derivazione delle relazioni che legano la potenza estratta ai parametri geometrici ed elettrici del trasduttore, quali numero di spire, resistenza di bobina, differenza di flusso tra posizione allineata e disallineata, e numero di denti della ruota, fino alla determinazione della potenza massima teoricamente trasferibile al carico e alla discussione delle condizioni di adattamento di impedenza in funzione della velocità di rotazione.

La parte centrale della tesi è dedicata alla progettazione dei circuiti di condizionamento della potenza per il sistema a sei fasi considerato. Vengono proposte e confrontate diverse topologie di raddrizzamento, valutate dapprima tramite simulazione circuitale e successivamente tramite la realizzazione di un banco di prova sperimentale, comprensivo della progettazione di una scheda PCB dedicata e della taratura dei sensori di corrente per la misura accurata della potenza estratta dal sistema. Infine, viene affrontata la scelta del circuito integrato di power management (PMIC) più adatto a regolare e immagazzinare l'energia raccolta, completando così la catena di conversione dall'energia meccanica rotazionale fino all'alimentazione stabile del carico finale.

Introduction

1.1 Motivation

The number of wireless sensor networks (WSNs) and Internet of Things (IoT) devices has grown rapidly in recent years, raising a fundamental question: how to power all these devices without replacing batteries or requiring constant maintenance. Recent advances in low-power electronics have reduced the energy consumption of sensor nodes to the range of tens to hundreds of microwatts [10]. At such low power levels, it becomes possible to power these devices by extracting energy from the surrounding environment, removing the need for batteries and allowing the network to operate indefinitely.

Batteries are the most common power source for WSN nodes, but they have well-known limitations. In many applications, such as structural health monitoring, industrial machinery surveillance, or medical implants, replacing or recharging depleted batteries is difficult or simply not possible [5]. Using larger batteries to extend the device lifetime increases the size of the system, which is often not acceptable. In addition, the environmental cost of battery and the consequent maintenance of large-scale networks are serious obstacles to the widespread use of WSNs [8].

Energy harvesting offers a practical solution to these problems. Instead of storing a fixed amount of energy, harvesting systems collect energy from sources available in the environment such as sunlight, heat gradients, mechanical vibrations, or radio frequency signals and convert it into electrical power to supply an electronic system.

Among all available energy sources, mechanical vibrations have received the most attention in the research community, accounting for more than half of

publications in this field [8]. Piezoelectric, triboelectric, and electromagnetic transducers are the most widely used mechanisms, each with different trade-offs in output voltage, power density, and complexity [10, 5]. Figure 1.1 shows the distribution of recent publications about transducer types, highlighting the dominance of mechanical energy harvesting approaches.

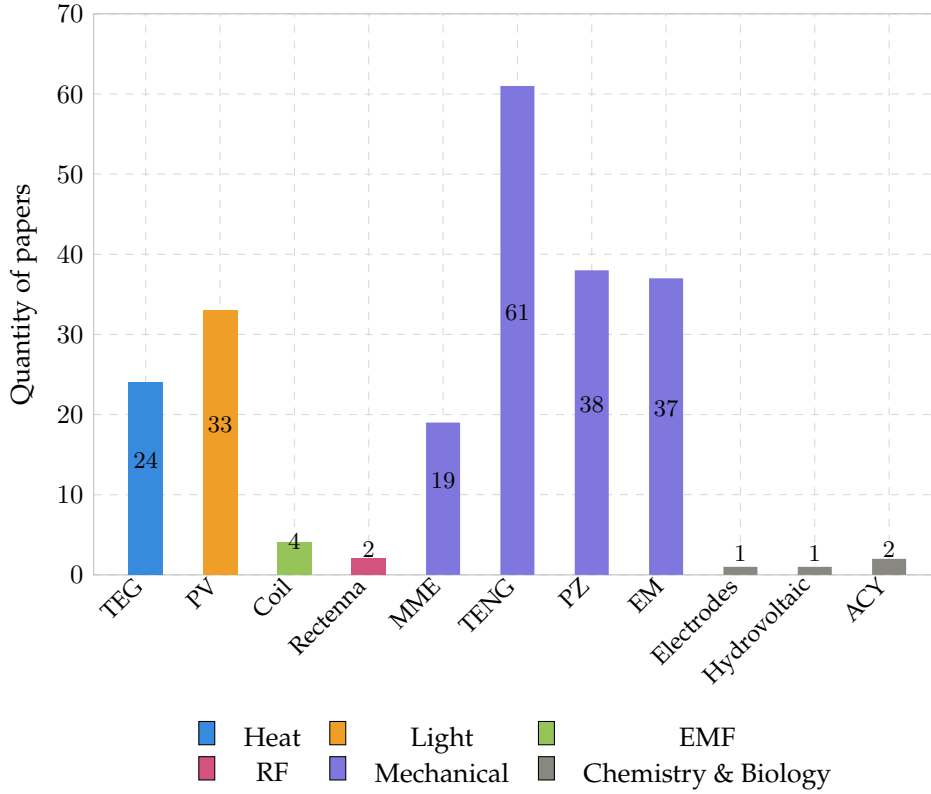


Figure 1.1: Distribution of papers based on transducer type [8].

Energy harvesting systems are generally divided into two categories [5]: systems that use the harvested energy immediately (*Harvest-Use*), and systems that store it first in a battery or supercapacitor before supplying it to the node (*Harvest-Store-Use*). The second approach is far more common, since most energy sources are not continuously available and some form of storage is needed to guarantee uninterrupted operation.

1.2 Energy Harvesting Systems

An energy harvesting system is made of three components that work in sequence, as illustrated in Figure 1.2.

- *Transducer*: converts the ambient energy into an electrical signal.
- *Power conditioning circuit*: perform AC-DC conversion, impedance matching and voltage regulation. A typical implementation combines a rectifier circuit with a power management integrated circuit (PMIC) [14].
- *energy storage element*: is used to keep the device operating for a certain amount of time even if the source is weak or no more present. Such element can be a capacitor/super capacitor or a rechargeable battery [10].

Figure 1.2: Block diagram of a typical energy harvesting system.

The performance of the entire system depends in particular on the transducer. Its design must be matched to the characteristics of the ambient source as well as to the power need of the target device. A transducer that is well suited to one environment may perform poorly in another, which is why the choice of transduction mechanism is the central design decision in any harvesting application [5].

Designing a good harvesting system is not just about getting the most power out of it. The system must work reliably in real conditions, where the energy source can be irregular or unpredictable. At the same time, it must be small, cheap, and easy to integrate into the target device [5]. Finding the right balance between these requirements is the main challenge in this field, and it has led to the development of many different solutions, as discussed in the following sections.

1.3 Performance metrics

A key challenge in evaluating energy harvesting systems is choosing the right performance metrics. Xu et al.[14] propose two metrics for this purpose. The first is the *power conversion efficiency* (PCE), which measures how much of the power entering the interface circuit actually reaches the output:

$$\text{PCE} = \eta_{\text{Rec}} \cdot \eta_{\text{DC-DC}} = \frac{P_{\text{out}}}{P_{\text{in}}}. \quad (1.1)$$

The second is the *power extraction efficiency* (PEE), which compares the output power to the maximum power the transducer could deliver under ideal conditions:

$$\text{PEE} = \frac{P_{\text{out}}}{P_{\text{max}}} = \frac{P_{\text{in}}}{P_{\text{max}}} \cdot \text{PCE}. \quad (1.2)$$

The difference between the two is important. A circuit can convert power efficiently (high PCE) while still wasting much of what the transducer could produce, if the circuit itself is not able to work in the optimal operating point. For this reason, PEE is a more useful metric when the goal is to maximise the total power delivered to the load [14].

1.4 Main Transduction Technologies

Mechanical vibrations can be converted into electrical energy through four main transduction mechanisms: piezoelectric, electromagnetic, electrostatic, and triboelectric. Each of them works in a different way and comes with its own advantages and limitations [9, 2].

Piezoelectric

Piezoelectric transducers produce a voltage when a material is mechanically deformed. The most common design is a cantilever beam with a piezoelectric layer on top: when the beam vibrates, the material deforms and generates an electric charge. This approach gives a high output voltage and does not need a magnet or moving parts, which makes it easy to integrate into small devices [1]. The main drawback is that piezoelectric materials are fragile and may crack under large deformations. Also, the output drops quickly when the vibration frequency moves away from the resonant frequency of the beam [10].

Electromagnetic

Electromagnetic transducers generate a voltage by moving a coil near a permanent magnet. This is the same principle used in large electrical generators, applied at a small scale. These harvesters tend to produce a low voltage but can deliver a relatively high current, which makes them a good choice when power output is the main goal. One limitation is that the coil and magnet must move relative to each other, which can be hard to achieve in applications involving large or slowly rotating objects [9].

Electrostatic

Electrostatic transducers work by changing the capacitance between two conductive plates as they move. When the plates get closer or further apart due to vibration, the stored electrical energy changes and can be extracted as useful power. This approach is easy to manufacture using standard microfabrication processes, making it attractive for very small devices. However, it requires an

external voltage or a pre-charged material to get started, which adds complexity to the system [9, 2].

Triboelectric

Triboelectric transducers use contact between two different materials to generate surface charges. When the materials touch and then separate, charges build up and can drive a current through an external circuit. This type of harvester, known as a triboelectric nanogenerator (TENG), was first demonstrated by Fan et al.[4] and has attracted a lot of interest because it can be made from cheap, flexible materials and works well at low frequencies, such as those produced by human motion. The main weaknesses are sensitivity to humidity and wear of the contact surfaces over time [8].

Among these four mechanisms, no single technology is universally superior. The best choice depends on the specific application, the characteristics of the available vibration source, and the constraints on size and cost. This comparison motivates the investigation of alternative approaches, including variable reluctance transduction, which is discussed in the following section.

1.5 The Variable Reluctance Principle

Variable reluctance energy harvesting (VREH) is a type of electromagnetic energy harvesting based on Faraday's law of induction. The transducer consists of three main parts: a permanent magnet, a pickup coil wound around a ferromagnetic pole piece, and a toothed ferromagnetic wheel that rotates nearby. When a tooth on the wheel aligns with the pole piece, the magnetic flux through the coil reaches its maximum value. When the gap between two teeth faces the pole piece, the flux drops to its minimum. This periodic variation in flux induces an alternating voltage across the coil, with a frequency that depends on the rotational speed and the number of teeth on the wheel [12]:

$$f_c = \omega N_t, \quad (1.3)$$

where ω is the rotational speed in rad/s and N_t is the number of teeth on the wheel. Following Faraday's law, the output power of the transducer delivered to a matched load can be expressed as [13]:

$$P_L = \frac{N_{\text{Coil}}^2}{8R_{\text{Coil}}} \Delta_\phi^2 \left[\frac{\omega N_{\text{Tooth}}}{2\pi} \right]^2, \quad (1.4)$$

where N_{Coil} is the number of coil turns, R_{Coil} is the coil resistance, $\Delta\phi$ is the magnetic flux difference between the aligned and unaligned positions, and N_{Tooth} is the number of teeth. This expression shows that the output power grows with the square of the rotational speed, and that it can be increased by maximising the flux difference and the number of coil turns, or by reducing the coil resistance.

What makes VREH different from most other electromagnetic harvesters is that neither the magnet nor the coil needs to move. The flux variation is caused entirely by the relative motion between the toothed wheel and the pickup units. This removes the need for a mechanical connection between the rotating part and the harvester, which is an advantage. It also means that the device is easy to scale: changing the shaft diameter mainly affects the size of the toothed wheel, while the pickup unit can stay the same [13]. In contrast, traditional electromagnetic generators require scaling of the magnets or coils, which becomes costly for large shafts or slow rotational speeds.

At low rotational speeds, the output voltage and frequency are both small, which makes the design of the interface circuit more critical [14]. At the same time, low-speed applications are exactly where VREH has the clearest advantage over other electromagnetic harvesters, since these typically require a relative displacement between coil and magnet that is hard to achieve at low speeds [9].

Its application to energy harvesting is more recent. Kroener et al. [7] were among the first to study a VREH specifically designed for energy harvesting in railway monitoring, showing that the principle could reliably power wireless sensor nodes from the motion of passing train wheels. Since then, the approach has been explored in a growing number of industrial and structural monitoring applications [12, 13].

1.6 State of the Art of VREH

The first applications of the variable reluctance principle to energy harvesting appeared in the context of railway monitoring. Kroener et al. [7] demonstrated that a VREH placed beside a rail track could generate enough power from the passage of train wheels to supply a wireless sensor node, reporting a peak output power of around 5.9 mW at a simulated train speed of 81.5 Km/h. This work showed for the first time that the variable reluctance principle could be practically used as an energy source rather than just a measurement transducer.

A systematic comparison of VREH topologies was later carried out by Xu et al. [12], who surveyed the structures proposed in the literature and evaluated them in terms of output power and power density for low-speed rotating applications. The study identified the *m-shaped* pickup unit, which uses two opposing

permanent magnets and a centre pole piece, as the topology with the highest volumetric power density among all structures considered. This result motivated a detailed design study of the m-shaped VREH, in which the influence of three key geometric parameters was investigated: the coil height h_{Coil} , the tooth height h_{Tooth} , and the magnet height h_{M} [13]. The study showed that these parameters involve trade-offs under size constraints, and provided a numerical optimisation method based on finite element analysis (FEA). The optimised structure reached output powers between 0.12 mW and 22 mW at rotational speeds of 5 rpm to 60 rpm within a volume constraint of 43.5 cm³, which is sufficient to power a range of duty-cycled wireless sensor systems [13].

A practical challenge in VREH systems is that the transducer produces a low-voltage alternating output, which must be rectified and conditioned before it can supply any systems. At low rotational speeds, the output voltage can be well below 1 V, which makes the choice of rectifier circuit critical. Xu et al. [14] compared three passive rectifier topologies: a full-wave bridge rectifier (FWR), a voltage doubler (VD), and a negative voltage converter (NVC), all on the same VREH system at speeds between 5 rpm and 20 rpm. The results showed that the VD achieved the best power extraction efficiency at low speeds, reaching a PEE of 40-60%, while the NVC outperformed it above 18 rpm. The FWR showed the lowest performance across all conditions, with PEE values of only 20-40%. These results confirm that the interface circuit is not just a passive element but actively affects how much power the transducer can deliver to the load.

These studies show that VREH is a mature and well-understood technology for low-speed rotational applications, with demonstrated power levels sufficient for a variety of wireless sensing applications. The most recent step forward in this direction is the multi-phase VREH proposed by Wu et al. [11], which integrates six m-shaped pickup units into a bearing hub unit for large commercial vehicles, reaching output powers between 0.46 W and 4.77 W at rotational speeds of 100 to 400 RPM. This significant increase in output power opens new possibilities for powering more demanding condition monitoring systems. However, it also introduces new challenges in the design of the power conditioning circuit, which must handle six independent AC outputs with different phases and varying voltage levels depending on the rotational speed. These challenges are discussed in the next section.

1.7 Open Challenges

Despite the progress described in the previous sections, several challenges remain open in the design of VREH systems. This section focuses on those that

are most relevant to the work presented in this thesis.

Power conditioning for multi-phase outputs

Single-phase VREH systems produce one alternating voltage signal that must be rectified and converted into a stable DC supply. This problem has been studied in detail for low-speed single-phase harvesters, where the choice of rectifier topology has a significant impact on the power extraction efficiency [14].

Many studies have investigated the design of power conditioning circuits for single-phase energy harvesters. The most common approach combines a rectification stage with a DC-DC converter, such as a Buck or Boost circuit, to regulate the output voltage to a level suitable for the target load [15, 16, 6, 3]. However, these solutions are designed for a single AC input and cannot be directly applied to a multi-phase harvester such as the MP-VREH, where six independent AC signals must be processed simultaneously.

Furthermore, many of these solutions do not include a startup circuit or a control system, which are both essential for the reliable operation of the power conditioning circuit in practice. Without a startup circuit, the system is not able to start operating when the harvested voltage is initially too low to power the control electronics. Without a control system, the circuit cannot adapt to changes in the rotational speed or load conditions, which are essential in real-world applications.

This gap in the literature motivates the investigation presented in this thesis.

When the number of pickup units increases, as in the multi-phase VREH proposed by Wu et al. [11], which uses six m-shaped pickup units arranged around a common toothed wheel, the power conditioning problem becomes considerably more complex. Each pickup unit produces an independent AC output with a different phase and the six signals are evenly spaced in time by 60° phase shift. This means that standard single-phase rectifier solutions cannot be directly applied, and that the interface circuit must be designed to handle multiple AC inputs efficiently.

Low output voltage at low speed

A fundamental limitation of VREH transducers is that the output voltage scales with rotational speed. At low speeds, the induced voltage may be well below 1 V, which makes rectification difficult because conventional diode bridges lose a significant fraction of the available voltage across their forward drops [14]. In a multi-phase system, this problem affects all six phases simultaneously. Finding a rectifier topology that works efficiently across a wide speed range, from the

minimum speed at which the harvester can sustain the load, up to the maximum operating speed, is therefore a key design requirement.

Impedance matching under variable conditions

The maximum power that a VREH can deliver to a load is achieved when the load impedance is matched to the internal impedance of the pickup coil. This matching condition depends on the rotational speed, because the electrical frequency of the output signal changes with speed. For a single-phase harvester, Xu et al. [13] showed that a fixed resistive load leads to an approximately linear increase in output power with speed, whereas a complex conjugate matched load, which compensates for the inductive component of the coil, could further increase the output power, especially at higher speeds. In a six-phase system, achieving dynamic impedance matching for all phases simultaneously adds a further layer of complexity to the interface circuit design.

1.8 Harvester Specifications and Research Objective

The work presented in this thesis is based directly on the MP-VREH proposed by Wu et al. [11]. The electrical characteristics of a single pickup unit, obtained from the transducer model, are summarised in Table 1.1. These values are used throughout this work as the reference operating conditions for the design and evaluation of the proposed power conditioning circuits.

RPM	f (Hz)	V_{Lpk} (V)	R_{coil} (Ω)	L_{coil} (mH)	P_o (W)
100	36.7	0.91	6	21.68	0.41
200	73.3	1.35	6	21.46	0.91
300	110.0	1.59	6	21.27	1.26
400	146.7	1.74	6	21.12	1.51

Table 1.1: Electrical parameters of a single MP-VREH pickup unit at different rotational speeds.

All values in Table 1.1 were obtained under impedance-matched load conditions. V_{Lpk} refers to the peak voltage across the load, and P_o is the average power dissipated in the load.

The overall system consists of six pickup units arranged in space so that they are electrically phase-shifted by 60° from one another.

These challenges motivate the work presented in this thesis, which focuses on the design and evaluation of power conditioning circuits for the six-phase MP-VREH described above. The goal is to identify the interface circuit topology

that maximises the power delivered to the load across the relevant operating speed range of 100 to 400 RPM (20-80 Km/h), corresponding to typical driving conditions of large commercial vehicles, for exaple buses.

Theoretical Background

This chapter introduces the physical and circuit-level concepts required to motivate the design choices made in the following chapters. In this chapter, starting from Faraday's law, an electrical model of the variable-reluctance energy harvester (VREH) is derived. The model is then used to discuss the maximum power transfer theorem and the role of impedance matching.

2.1 Electrical Model of the Variable-Reluctance Harvester

2.1.a Faraday's law

The Faraday's law is the starting point for any electromagnetic transducer, which relates the electromotive force $\varepsilon(t)$ (EMF) induced in a closed loop wire to the time rate of change of the magnetic flux $\Phi(t)$ linked to the loop.

$$\varepsilon(t) = -\frac{d\Phi(t)}{dt} \quad (2.1)$$

For a coil made by N_t turns sharing the same flux, the total linked flux is $\Lambda(t) = N_t\Phi(t)$ and the induced EMF becomes:

$$\varepsilon(t) = -\frac{d\Lambda(t)}{dt} = -N_t \frac{d\Phi(t)}{dt}. \quad (2.2)$$

In case of a coil consisting of N_t turns mechanically spaced one respect to each other leading to different linked flux, the induce voltage can be described by

$$\varepsilon(t) = -\frac{d}{dt} \cdot \sum_{n=i}^{N_t} \Phi_i(t) \quad (2.3)$$

where Φ_i is the flux linkage associated to a single turn of the coil [12].

2.1.b Variable Reluctance Principle

Unlike conventional electromagnetic transducers, in which the EMF is induced by a relative motion between a permanent magnet (PM) and a pickup coil, a variable-reluctance energy harvester (VREH) keeps both the magnet and the coil stationary. The magnetic flux linked to the coil is instead modulated by the passage of a rotating, toothed ferromagnetic wheel in front of a stationary pickup unit: as a tooth or a slot alternately faces the pickup unit, the reluctance of the magnetic circuit changes periodically, and it is this reluctance variation (rather than any motion of the magnet) that drives the flux change $\Phi_i(t)$ in (2.3) [12]. Because the rotating part carries no magnet, no coil and no electrical connection, the resulting transducer is mechanically simple, robust, and easily scalable to shafts of different diameters [13].

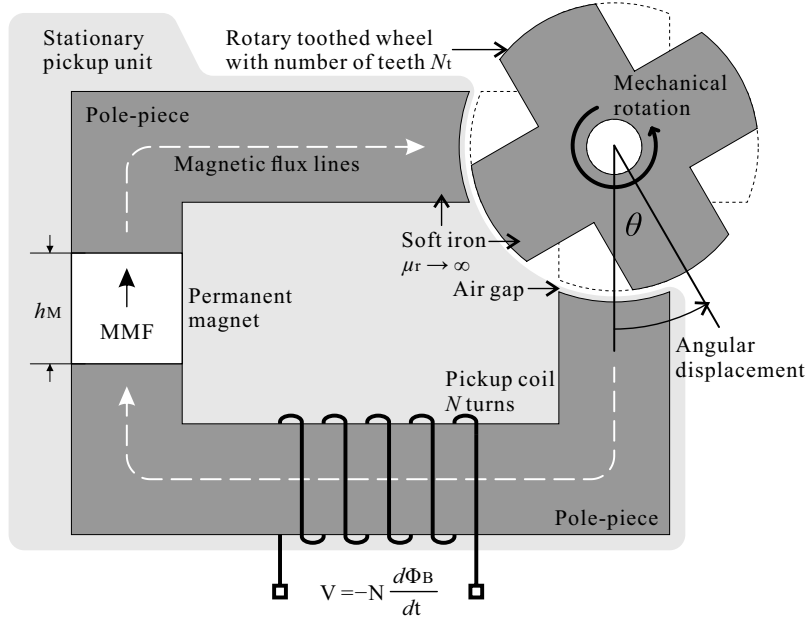


Figure 2.1: Structure of a variable-reluctance sensor: a stationary pickup unit, composed of a permanent magnet, a soft-ferromagnetic pole-piece and a pickup coil, faces a rotating toothed wheel across an air gap.

The pole-piece and the toothed wheel are both made of soft ferromagnetic material, while the PM is a hard ferromagnetic material providing a magnetomotive force (MMF) $\mathcal{F}_M$ that can be taken as constant. As the wheel rotates, the pickup unit cycles between two extreme relative positions: the *aligned* position,

where a pole of the pickup unit faces a tooth of the wheel and the air-gap reluctance is minimum, and the *unaligned* position, where the pole faces a slot and the reluctance is maximum [12, 13]. The corresponding flux swings between a maximum Φ_{max} and a minimum Φ_{min} , so that the relevant design quantity is the flux difference $\Delta\Phi = \Phi_{max} - \Phi_{min}$ between the two extremes. Fig. 2.2 illustrates this swing directly: the air-gap flux density rises sharply to a plateau at the aligned position and nearly collapses at the unaligned position, confirming that the flux change is concentrated within a narrow angular window around each tooth-slot transition.

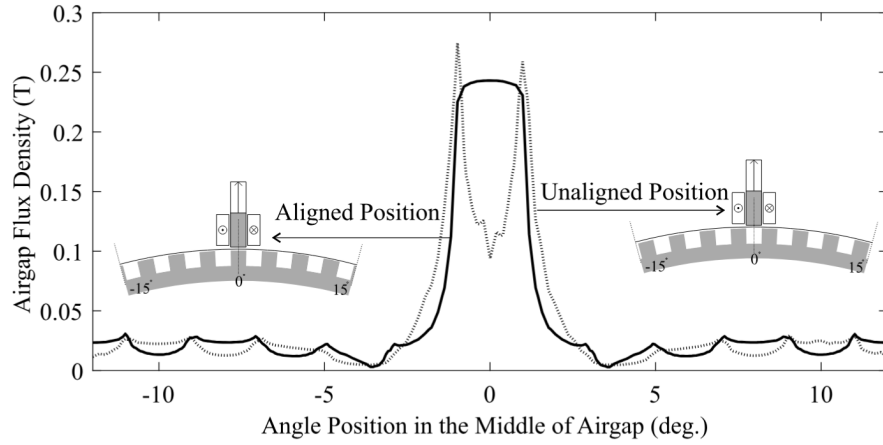


Figure 2.2: Airgap flux density distribution for aligned and unaligned positions. Reproduced from [12].

From flux swing to output power

For a wheel with N_t teeth rotating at angular velocity ω , the electrical frequency of the induced voltage is

$$f_c = \omega N_t. \quad (2.4)$$

At low rotational speeds the coil impedance is dominated by its resistance R_{Coil} , so that maximum power is transferred to a purely resistive load matched to it, $R_L = R_{Coil}$; under this condition the load voltage $V_L(t)$ is related to the open-circuit coil voltage $V_c(t)$ of (2.3) by

$$V_L(t) = \frac{1}{2} V_c(t). \quad (2.5)$$

Assuming the time-varying flux to be an ideal sinusoidal wave, uniformly distributed in the N_c turns of the coil, the peak-to-peak voltage developed across

the matched load follows from (2.3) and (2.5) as [12]

$$V_{L,pp} = -\frac{1}{2} N_c \frac{\Phi_{max} - \Phi_{min}}{\frac{1}{2}T_c} = -N_c f_c \Delta\Phi, \quad (2.6)$$

where $\frac{1}{2}T_c$ is the time taken by the wheel to rotate from the aligned to the unaligned position, i.e. from a tooth to a slot centred on the pole-piece.

Because $V_{L,pp}$ in (2.6) is the peak-to-peak amplitude of an (assumed) sinusoidal load voltage, its root-mean-square value is $V_{L,rms} = V_{L,pp}/(2\sqrt{2})$, so that the RMS power dissipated in the matched load is [12]

$$P_L = \frac{V_{L,rms}^2}{R_L} = \frac{V_{L,pp}^2}{8 R_{Coil}} = \frac{(\omega N_t N_c)^2}{8 R_{Coil}} (\Delta\Phi)^2, \quad (2.7)$$

where the last equality follows from substituting (2.4) into (2.6). Equation (2.7) shows that the harvested power scales quadratically with the rotational speed ω , the flux swing $\Delta\Phi$, the number of coil turns N_c and the number of teeth N_t , while it is inversely proportional to the coil resistance R_{Coil} .

For a given application (i.e. fixed ω and N_t), the design of the pickup unit therefore splits into two largely independent optimization problems [13]:

$$P_L \propto \frac{N_c^2}{R_{Coil}}, \quad (2.8a)$$

$$P_L \propto (\Delta\Phi)^2, \quad (2.8b)$$

Sub-function (2.8a) is a winding-design problem: for a coil of resistivity ρ_{wire} occupying a cross-section of height h_{Coil} and width W_{Coil} inside the pole-piece, it can be shown that (2.8a) reduces to [13]

$$P_L \propto \frac{h_{Coil} W_{Coil} k_w}{\rho_{wire}}, \quad (2.9)$$

with $k_w < 1$ the winding fill factor, defined as the fraction of the winding-window cross-section $h_{Coil} W_{Coil}$ actually occupied by copper [13]. For a fixed available cross-section, output power is thus maximized simply by maximizing the copper fill of the winding window. Sub-function (2.8b), in contrast, is a magnetic-circuit design problem. Once the magnet is chosen, its coercivity H_c and height h_M fix the MMF $\mathcal{F}_M = H_c h_M$, so $\Delta\Phi$ can no longer be increased by acting on the magnet alone. What remains under the designer's control is the shape of the pole-piece and toothed wheel around the air gap, thus the reluctance should be made as small as possible at the aligned position and as large as possible at the unaligned position, so that the two extremes are as far

apart as possible and $\Delta\Phi$ is maximized. This magnetic-circuit optimization is detailed next.

Magnetic-circuit model of the reluctance swing

To turn $\Delta\Phi$ into a design formula, the magnetic circuit can be treated like an electrical one, with reluctance playing the role of resistance. The reluctance of a simple flux-path element of length l , cross-section A and relative permeability μ_r is

$$\mathcal{R} = \frac{l}{\mu_0 \mu_r A}. \quad (2.10)$$

The pole-piece and the toothed wheel are made of soft ferromagnetic material, whose relative permeability is several orders of magnitude larger than that of air ($\mu_r \approx 1$) or of the PM ($\mu_r \approx 1.05$). Since a larger μ_r means a much smaller reluctance, the pole-piece and the wheel contribute essentially nothing to the total loop reluctance [13]. What is left, at either rotor position, is just two terms in series: the air-gap reluctance and the magnet's own reluctance,

$$\mathcal{R}_M = \frac{h_M}{\mu_0 \mu_M A_M}. \quad (2.11)$$

Denoting by $\mathcal{R}_{g,a}$ and $\mathcal{R}_{g,u}$ the air-gap reluctance at the aligned and unaligned positions, Hopkinson's law gives the flux at each extreme simply as the MMF divided by the loop reluctance, $\Phi_{max} = \mathcal{F}_M / (\mathcal{R}_{g,a} + \mathcal{R}_M)$ and $\Phi_{min} = \mathcal{F}_M / (\mathcal{R}_{g,u} + \mathcal{R}_M)$. Subtracting the two gives the flux swing already introduced in (2.8b), now written out explicitly:

$$\Delta\Phi = \mathcal{F}_M \left(\frac{1}{\mathcal{R}_{g,a} + \mathcal{R}_M} - \frac{1}{\mathcal{R}_{g,u} + \mathcal{R}_M} \right). \quad (2.12)$$

Once the magnet material and the nominal air-gap length are fixed by mechanical constraints, (2.12) shows that only two geometric parameters are left to control $\Delta\Phi$:

- the **tooth height** h_{Tooth} , which affects only $\mathcal{R}_{g,u}$: a taller tooth raises the reluctance seen at the unaligned position, but this benefit saturates once the tooth is taller than about 3 mm, while the aligned-position reluctance $\mathcal{R}_{g,a}$ stays essentially unchanged [13];
- the **magnet height** h_M , which pulls in two opposite directions at once: it raises the MMF $\mathcal{F}_M = H_c h_M$ (good for $\Delta\Phi$), but it also raises the magnet's own reluctance $\mathcal{R}_M$ (bad for $\Delta\Phi$). Because of this trade-off, a non-trivial optimal value of h_M exists.

In a size-constrained design, what matters is not just P_L but the power delivered per unit volume, i.e. the volumetric optimization of the pickup unit. Modelling the harvester as a hollow cylinder of overall diameter d_o , shaft diameter d_{Shaft} and depth l_o , the volumetric power density is

$$p_v = \frac{P_L}{V_{VREH}}, \quad V_{VREH} = \frac{\pi}{4} l_o (d_o - d_{Shaft})^2. \quad (2.13)$$

Once the air-gap length and the (shared) flux-path width are fixed, the available radial space must be split between the coil and the tooth, $C = h_{Coil} + h_{Tooth}$. Growing h_{Coil} always helps P_L , through (2.9), whereas growing h_{Tooth} only helps up to the 3 mm knee point described above, through (2.12). Because of this asymmetry, for any given C there is an optimal split between h_{Coil} and h_{Tooth} that maximizes p_v [13].

The same reasoning applies to the magnet height by setting $\frac{d\Delta\Phi}{dh_M} = 0$ in (2.12) yields a closed-form expression for the optimal magnet height,

$$h_M^* = \mu_M \frac{k_2}{k_1} \frac{W_{g,a}}{W_{g,u}} \sqrt{l_{g,a} l_{g,u}}, \quad (2.14)$$

where $l_{g,a}$, $l_{g,u}$, $W_{g,a}$, $W_{g,u}$ are the effective air-gap length and width at the aligned and unaligned positions, and $k_1, k_2 > 1$ are correction factors that account, respectively, for flux leakage and for the MMF drop across the finite-permeability soft iron [13].

2.1.c Electrical Thevenin Model of the Pickup Unit

The Faraday law derivation of Section 2.1.a gives the ideal electromotive force induced in the coil, but it does not yet describe how that voltage appears at the physical terminals of the pickup unit. A real coil is made of a finite length of copper wire, so it has a nonzero winding resistance, and it carries a current that produces its own magnetic flux, so it also has a nonzero self inductance. Both effects are in series with the ideal source, since the same wire that generates the electromotive force also carries the resistance and the flux linkage that define R_{Coil} and L_{Coil} .

Circuit equation

Let $e(t)$ denote the ideal electromotive force predicted by (2.3), and let $i(t)$ be the current flowing out of the coil into the external circuit. Applying Kirchhoff's voltage law around the loop formed by the source, the winding resistance, the

winding inductance and the external circuit gives

$$v(t) = R_{Coil} i(t) + L_{Coil} \frac{di(t)}{dt} + e(t), \quad (2.15)$$

where $v(t)$ is the voltage measured at the coil terminals, available to the power conditioning circuit. In other words, (2.15) models the pickup unit as an ideal voltage source $e(t)$ in series with a resistance R_{Coil} and an inductance L_{Coil} , feeding the external circuit at the coil terminals; this series generator model is used throughout the rest of this thesis.

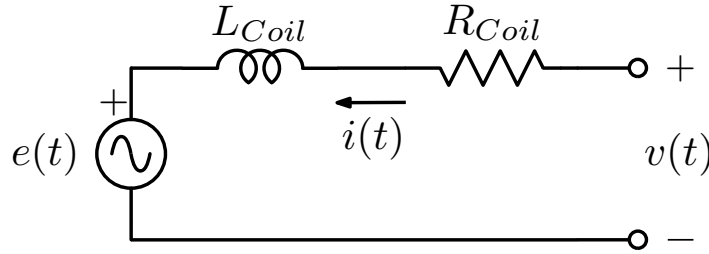


Figure 2.3: Series generator model of the pickup unit: an ideal source in series with the winding resistance R_{Coil} and inductance L_{Coil}

Winding resistance

The winding resistance follows directly from the resistivity of the wire, its total length and its cross sectional area,

$$R_{Coil} = \rho_{wire} \frac{N_c l_t}{A_w}, \quad (2.16)$$

where N_c is the number of turns, l_t is the mean length of a single turn, and A_w is the cross sectional area of the wire. If the coil occupies a winding window of height h_{Coil} and width W_{Coil} with a copper fill factor k_w , the wire cross section can be written as $A_w = k_w h_{Coil} W_{Coil} / N_c$, so that (2.16) becomes

$$R_{Coil} = \rho_{wire} \frac{N_c^2 l_t}{k_w h_{Coil} W_{Coil}}. \quad (2.17)$$

Treating l_t as approximately constant for a fixed pole piece geometry, (2.17) reproduces the proportionality $N_c^2 / R_{Coil} \propto k_w h_{Coil} W_{Coil} / \rho_{wire}$ already used in (2.9) to optimize the winding cross section [13].

Self inductance and its dependence on rotor position

The self inductance of a coil linked to a magnetic circuit of total reluctance $\mathcal{R}$ is, by the same reluctance used for Hopkinson's law in (2.10),

$$L_{Coil} = \frac{N_c^2}{\mathcal{R}}. \quad (2.18)$$

The pickup unit is a variable-reluctance device, so the reluctance seen by the coil's own flux changes with the rotor angle too, in the same way as the reluctance that shapes the induced EMF. This means the self inductance is not constant: it swings between two extremes,

$$L_{max} = \frac{N_c^2}{\mathcal{R}_{g,a} + \mathcal{R}_M}, \quad L_{min} = \frac{N_c^2}{\mathcal{R}_{g,u} + \mathcal{R}_M}, \quad (2.19)$$

reached at the aligned and unaligned positions, just like the flux extremes of (2.12). For circuit-level analysis, such as the impedance matching study of Section 2.1.d, it is more practical to replace this varying inductance with a single, constant value L_{Coil} . This value can be obtained either by averaging over one electrical period, or by measuring it directly with an impedance analyzer at the frequency of interest. It is this constant value, $L_{Coil} \approx 210$ mH, that is used later in this chapter.

Phasor representation

For a sinusoidal steady state at the electrical frequency f_c of (2.4), the time domain relation (2.15) can be written in phasor form as

$$V = I(R_{Coil} + j\omega_e L_{Coil}) + E, \quad (2.20)$$

where E , I and V are the phasors associated with $e(t)$, $i(t)$ and $v(t)$, and $\omega_e = 2\pi f_c$. Equation (2.20) shows that, seen from its electrical terminals, the pickup unit behaves as an ideal source E in series with the impedance

$$Z_s = R_{Coil} + j\omega_e L_{Coil}, \quad (2.21)$$

which is precisely the Thévenin source impedance used in (2.22) for the impedance matching analysis of Section 2.1.d. The series generator, resistance, inductance model derived in this section therefore provides the electrical foundation on which the maximum power transfer condition of the next section, and the power conditioning topologies compared in Chapter 3, are built.

2.1.d Maximum Power Transfer and Impedance Matching

The previous section modelled the pickup unit as a Thévenin source, an open circuit EMF $e(t)$ in series with the coil's own impedance. At low electrical frequency this impedance is essentially the winding resistance R_{Coil} , and matching a resistive load to it, $R_L = R_{Coil}$, is enough to extract the maximum available power, as already used in (2.7). As the rotational speed increases, however, the coil inductance L_{Coil} can no longer be neglected, and the condition for maximum power transfer must be generalized. This section derives that general condition and explains why it becomes a central design constraint for a harvester that must operate between 100 and 400 rpm (the target range of this thesis).

2.1.e Thévenin model of the source and the load

Let the coil be represented, at the electrical frequency of interest, by its Thévenin equivalent, an RMS open circuit phasor voltage V_c in series with the source impedance

$$Z_s = R_{Coil} + j\omega_e L_{Coil}, \quad (2.22)$$

where $\omega_e = 2\pi f_c$ is the electrical angular frequency and f_c is the electrical frequency defined in (2.4). The load presented to the coil by the power conditioning circuit is likewise represented, at the same frequency, by a generic complex impedance

$$Z_L = R_L + jX_L. \quad (2.23)$$

Here R_L is the resistive part of the load and X_L its reactive part, which can be inductive ($X_L > 0$) or capacitive ($X_L < 0$) depending on how the load is realized.

2.1.f Power delivered to the load

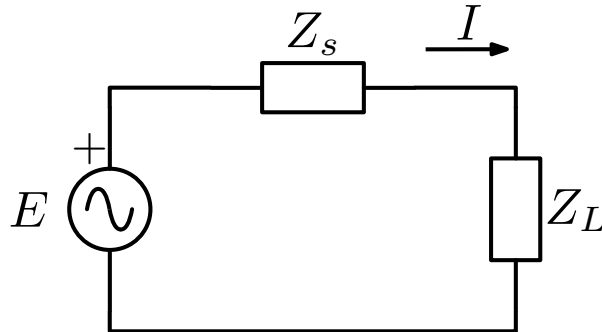


Figure 2.4: Thévenin equivalent circuit used for the impedance matching analysis: source E with source impedance Z_s , driving a generic load impedance Z_L .

With Z_s and Z_L in series, the RMS current flowing in the loop is

$$I = \frac{E}{Z_s + Z_L}, \quad (2.24)$$

and the average power dissipated in the resistive part of the load is

$$P_L = |I|^2 R_L = \frac{|E|^2 R_L}{(R_{Coil} + R_L)^2 + (\omega_e L_{Coil} + X_L)^2}. \quad (2.25)$$

Equation (2.25) is the generalization of (2.7) to a complex load. Setting $L_{Coil} = 0$ and $X_L = 0$ recovers the purely resistive case already used earlier in this chapter.

2.1.g The complex conjugate matching condition

Equation (2.25) can be maximized in two independent steps, since R_L and X_L appear in separate terms of the denominator.

First, for any fixed R_L , the denominator is minimized, and P_L is therefore maximized, by cancelling the reactive term. This requires

$$X_L = -\omega_e L_{Coil}. \quad (2.26)$$

With this choice, (2.25) reduces to

$$P_L = \frac{|E|^2 R_L}{(R_{Coil} + R_L)^2}. \quad (2.27)$$

Second, differentiating (2.27) with respect to R_L and setting the result to zero,

$$\frac{\partial P_L}{\partial R_L} = |E|^2 \frac{R_{Coil} - R_L}{(R_{Coil} + R_L)^3} = 0 \implies R_L = R_{Coil}. \quad (2.28)$$

Combining (2.26) and (2.28) gives the well known maximum power transfer theorem in its general, complex form,

$$Z_L = Z_s^* = R_{Coil} - j\omega_e L_{Coil}. \quad (2.29)$$

The load must present a resistance equal to the coil resistance and a reactance equal in magnitude but opposite in sign to the coil reactance. Physically, the load must absorb energy from the coil (resistive part) while returning, every half cycle, exactly the reactive energy that the coil inductance would otherwise store (capacitive part). Substituting (2.29) into (2.25) gives the maximum available

power,

$$P_{L,\max} = \frac{|E|^2}{4 R_{Coil}}, \quad (2.30)$$

which is independent of L_{Coil} . Once the reactance is properly cancelled, the maximum extractable power depends only on the coil resistance, exactly as in the low frequency case of (2.7).

2.1.h Realizing the reactive compensation

Condition (2.26) shows that an inductive coil requires a load with an equal and opposite *capacitive* reactance. The simplest way to obtain this is a compensation capacitor C_m placed in series with the resistive load, whose reactance at ω_e is $X_{C_m} = -1/(\omega_e C_m)$. Imposing $X_{C_m} = -\omega_e L_{Coil}$ gives the required capacitance

$$C_m = \frac{1}{\omega_e^2 L_{Coil}}. \quad (2.31)$$

Once the capacitor C_m is fixed according to (2.31), it cancels the coil's reactance and therefore restores the resistive matching condition of (2.28) and the maximum power of (2.30) only at the single electrical frequency ω_e for which it was chosen.

2.2 Multi-Phase System Model

2.2.a System Overview

The harvester considered in this thesis is the six-phase multi-phase variable-reluctance energy harvester (MP-VREH) proposed by Wu et al. [11] and already introduced in Section 1.8, whose electrical specifications are summarised in Table 1.1. It consists of six m-shaped pickup units, each electrically equivalent to the series generator derived in Section 2.1.c, arranged around a common toothed wheel and integrated into a single bearing hub. As the wheel rotates, each pickup unit produces an independent AC voltage at the electrical frequency $f_c = \omega N_t$ of (2.4). The six pickup units are evenly spaced around the wheel, so that their induced voltages are mutually phase-shifted by 60° , one from the next.

From an electrical standpoint, the harvester can therefore be represented as six identical Thévenin sources, each with open-circuit voltage $e_k(t)$, series resistance R_{Coil} and series inductance L_{Coil} , whose only difference is a fixed phase offset. This six-source representation is the starting point for the power conditioning circuits developed in Chapter 3, which must process six independent AC inputs rather than a single one.

2.2.b Multi-Phase Electrical Model

Let $\omega_e = 2\pi f_c$ be the electrical angular frequency common to all six phases, and let $k = 0, 1, \dots, 5$ index the pickup units in the order in which they are met by a tooth of the wheel. Under the ideal assumption of even angular spacing, the open-circuit EMF of phase k is a sinusoid of amplitude E_{pk} delayed with respect to phase 0 by $k \cdot 60^\circ$,

$$e_k(t) = E_{pk} \sin\left(\omega_e t - k \frac{\pi}{3}\right), \quad k = 0, 1, \dots, 5. \quad (2.32)$$

Each phase drives the power conditioning circuit through its own series impedance $Z_s = R_{Coil} + j\omega_e L_{Coil}$, exactly as derived for the single pickup unit in (2.22). Because R_{Coil} , L_{Coil} and E_{pk} are the same for all six units, the six branches differ only in the phase term of (2.32); so the same circuit design can be reused for every phase, each one simply delayed in time by 60° .

2.2.c Design Requirements for the Interface Circuit

The specifications summarised in Table 1.1 fix the operating envelope that the interface circuit must handle. The load voltages V_{Lpk} reported there are measured under the impedance-matched condition of (2.5), i.e. half of the open-circuit source amplitude E_{pk} of (2.32) used throughout this section; over the 100–400 rpm target speed range, this puts E_{pk} at about 1.8 V to 3.5 V. The rectifier must remain efficient at these low input voltages. Because the six phases are electrically identical but time-shifted, as shown in (2.32), the interface circuit must either replicate the same rectification stage six times, one per phase, or combine the six inputs in a way that preserves the impedance matching condition of Section 2.1.d for each of them individually. These two requirements, low-voltage operation and per-phase impedance matching across a variable speed range, motivate the comparison of rectifying topologies carried out in Chapter 3.

Proposed Rectifying Circuits

3.1 Overview

Section 1.8 showed that, over the target speed range, the peak voltage available at each phase of the MP-VREH stays low, which makes the choice of rectifier circuit critical. Implementing a design that wastes a large, fixed voltage drop at every conduction cycle can lose a substantial fraction of the already limited available power. Xu et al. [14] compared three passive rectifier topologies for this purpose, a full-wave bridge rectifier (FWR), a voltage doubler (VD) and a negative voltage converter (NVC), using the PCE and PEE metrics of (1.1) and (1.2), and found that no single topology is best across the whole speed range. This chapter focuses on the two topologies at the extremes of that comparison, the FWR, as the simplest and cheapest baseline, and the NVC, as the best performer at higher speed, and discusses how each one must be adapted to rectify the six phase-shifted inputs of (2.32) rather than a single-phase source.

Rather than rectifying each of the six phases independently, the six inputs can be grouped so as to reuse existing, well-documented multi-phase rectifier bridges. For the FWR, this thesis groups the six phases into two three-phase subsystems, each rectified by a three-phase bridge, a topology already well documented in the literature. For the NVC, which was identified above as the best performer at higher speed, the six phases are instead grouped into three two-phase subsystems, each pairing two phases that are 180° apart and connecting them in anti-series so that their voltages add instead of cancelling. These two grouping schemes are described in detail in the dedicated sections of this chapter and in Chapter 4.

3.2 Full-Wave Rectifier (FWR)

Figure 3.1 shows the three-phase bridge used to rectify one of the two three-phase groups introduced in the overview of this chapter. Taking every other pickup unit of (2.32) (phases $k = 0, 2, 4$, or equivalently $k = 1, 3, 5$) gives exactly such a group: three sources 120° apart, the spacing a three-phase bridge rectifier is designed for. The three input terminals are labelled A, B and C in the figure. Diodes D_1, D_3 and D_5 share a common cathode at OUT+, with their anodes tied to A, B and C respectively; diodes D_2, D_4 and D_6 share a common anode at OUT-, with their cathodes tied to A, B and C respectively.

At any instant, only the top diode connected to whichever of the three phases is momentarily the highest is forward biased, conducting that phase's current to OUT+; likewise, only the bottom diode connected to whichever phase is momentarily the lowest is forward biased, returning current from OUT- to that phase. The remaining four diodes are reverse biased and block. The current path therefore always includes exactly two diodes in series, one from the top group and one from the bottom group, so the voltage across OUT+ and OUT- is always lower than the instantaneous difference between the highest and lowest phase voltages by two forward voltage drops. A filter capacitor at the output reduces the ripple left by this waveform.

3.2.a Star or Delta Connection of the Three-Phase Group

Figure 3.1 fixes how A, B and C are rectified, but not how the three independent pickup coils of the group are wired together to produce those three terminals in the first place. Two configurations are considered for a group of phases $k, k + 2, k + 4$.

In a *star* (Y) connection, one terminal of each of the three coils is joined at a common internal node, left floating and not brought out to the bridge; the other, free terminal of each coil becomes A, B or C . Denoting the floating node voltage $v_S(t)$, the terminal voltages are $v_A(t) = v_S(t) + e_k(t)$, $v_B(t) = v_S(t) + e_{k+2}(t)$ and $v_C(t) = v_S(t) + e_{k+4}(t)$; the line-to-line voltage seen by the bridge does not depend on $v_S(t)$, since it cancels in the difference,

$$v_A(t) - v_B(t) = e_k(t) - e_{k+2}(t) = \sqrt{3} E_{pk} \cos\left(\omega_e t - k \frac{\pi}{3} - \frac{\pi}{3}\right), \quad (3.1)$$

an amplitude $\sqrt{3}$ times larger than a single phase, and leading it by 30° .

In a *delta* (Δ) connection, the three coils are instead connected end to end in a closed loop, so that each pair of terminals A - B , B - C , C - A coincides exactly with the two terminals of one coil. The line-to-line voltage is then simply the coil's

own EMF,

$$v_A(t) - v_B(t) = e_k(t), \quad (3.2)$$

with no amplitude increase. The closed loop must carry the sum of the three EMFs, which is identically zero for a balanced set ($e_k(t) + e_{k+2}(t) + e_{k+4}(t) = 0$ for any t , since the three phases are 120° apart), so no circulating current flows around the loop from the sources alone.

The star connection therefore delivers a $\sqrt{3}$ times larger voltage to the bridge for the same pickup units, which helps against the low-voltage operation discussed below, at the cost of an inaccessible internal node whose potential is fixed only by the balance of the three branch currents. The delta connection avoids this floating node, at the cost of losing the $\sqrt{3}$ gain. Both configurations are carried forward as candidates for the FWR stage in the simulations of Chapter 4.

Because the two-diode drop of the FWR is essentially constant regardless of the phase voltages, it removes a larger fraction of the signal the lower those voltages are. Given the phase peak voltage range of Section 1.8, the MP-VREH operates exactly in this low-voltage regime, where the fixed diode drop is expected to dominate the losses of the FWR stage; the star connection of (3.1) mitigates this by a factor of $\sqrt{3}$, while the delta connection of (3.2) does not. Its main advantage remains simplicity: the FWR needs no active devices, no gate drive and no threshold voltage to overcome, which makes it a useful baseline against which the NVC of Section 3.3 is compared.

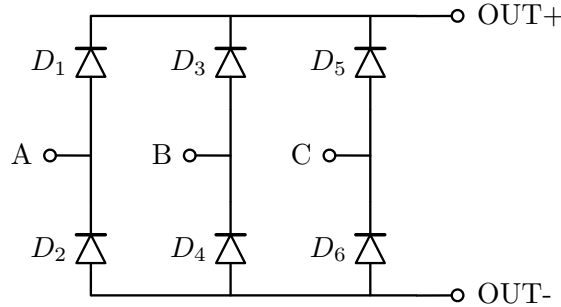


Figure 3.1: Three-phase full-wave bridge rectifier used for one of the two three-phase groups of pickup units, with input terminals A , B , C .

3.3 Negative Voltage Converter (NVC)

Figure 3.2 shows the two-phase NVC used to rectify one of the three two-phase groups introduced in the overview of this chapter. Rather than pairing adjacent phases, each group combines two pickup units that are electrically in antiphase:

from (2.32), phases k and $k + 3$ satisfy

$$e_{k+3}(t) = E_{pk} \sin\left(\omega_e t - (k + 3) \frac{\pi}{3}\right) = E_{pk} \sin\left(\omega_e t - k \frac{\pi}{3} - \pi\right) = -e_k(t), \quad (3.3)$$

since they are 180° apart. Connecting the two coils in anti-series, i.e. joining their like-polarity terminals together and leaving the other two terminals free, gives a combined open-circuit voltage across the free terminals A and B of

$$v_A(t) - v_B(t) = e_k(t) - e_{k+3}(t) \stackrel{(3.3)}{=} 2e_k(t), \quad (3.4)$$

twice the amplitude of a single phase, at the cost of a doubled series impedance $2Z_s$. Connecting the same pair in straight series, by contrast, would simply cancel the two antiphase voltages to zero, since $e_k(t) + e_{k+3}(t) = 0$. The three NVC groups therefore pair phases $k = 0, 3$; $k = 1, 4$; and $k = 2, 5$, each combined in anti-series at the terminals A and B of Figure 3.2. The bridge replaces the diodes of the FWR with four MOSFETs S_1 – S_4 , each with a Schottky diode (D_1 – D_4) connected in parallel. S_1 and S_2 connect terminal A to OUT+ and OUT- respectively, while S_3 and S_4 connect terminal B to OUT+ and OUT- respectively. The bridge is *gate cross-coupled*: the gates of S_1 and S_2 are tied together and driven directly by terminal B , while the gates of S_3 and S_4 are tied together and driven directly by terminal A . No separate gate-drive or polarity-detection circuit is needed, since each pair of switches is driven by the very AC terminal it is not connected to.

When A is higher than B , the gate voltage B shared by S_1 and S_2 turns S_1 on and keeps S_2 off, while the gate voltage A shared by S_3 and S_4 turns S_4 on and keeps S_3 off. The load current therefore flows from A through S_1 to OUT+, and returns from OUT- through S_4 to B . When B is higher than A , the roles swap, and the current instead flows through S_3 and S_2 . In either case, the forward voltage drop across the bridge is the sum of the drain-source voltages of the two conducting MOSFETs, rather than a fixed diode drop as in the FWR.

This switching mechanism only works once the gate-source voltage across the conducting pair is large enough to turn the MOSFETs fully on. Below a critical voltage difference $V_{\text{off}} \approx 1.2$ V between A and B , set by the largest threshold voltage among the four MOSFETs, the devices cannot switch properly, and the parallel Schottky diodes D_1 – D_4 carry the current instead: in this regime the NVC behaves like an FWR, with the same two-diode-drop loss. Above V_{off} , the MOSFETs conduct with an on-resistance much smaller than a diode's forward drop, and the output voltage tracks the input with very low losses.

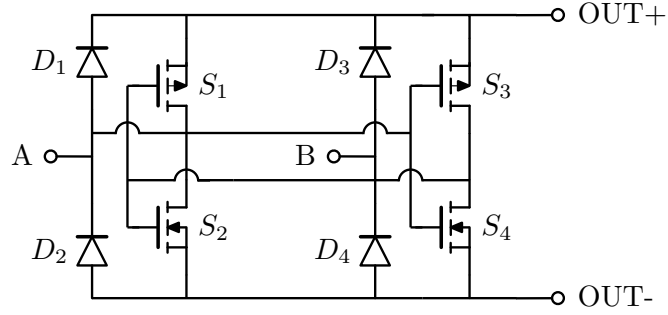


Figure 3.2: Negative voltage converter used for one of the three two-phase groups of pickup units, with input terminals A , B .

3.4 Application to the Six-Phase System

Sections 3.2 and 3.3 already present the FWR and the NVC in the multi-phase form adopted in this thesis. The FWR groups the six pickup units of Section 2.2 into two three-phase subsystems, taking phases $k = 0, 2, 4$ and $k = 1, 3, 5$ of (2.32) respectively. The NVC, instead, groups them into three two-phase subsystems, each pairing two phases 180° apart and connecting them in anti-series, as derived in (3.4) ($k = 0, 3$; $k = 1, 4$; $k = 2, 5$). Because all six phases share the same R_{Coil} , L_{Coil} and peak voltage and differ only by the 60° time shift, every group of a given topology sees identical operating conditions, only shifted in time, so the performance of one group is representative of all the others. This is the grouping carried forward into the simulations of Chapter 4.

4

Simulations

Experimental Measurement Setup

5.1 Overview

The circuit simulations of Chapter 4 predict the behaviour of the FWR and NVC stages introduced in Chapter 3, but validating those predictions requires measuring the real prototype. Both the voltage and the current at each phase, and at the output of each rectifying stage, must be acquired at the same time in order to compute the electrical power. This chapter describes the measurement setup built for this purpose, from the acquisition instrument down to the current sensing elements and their calibration.

Two precision instruments are used throughout this chapter wherever a known reference quantity, rather than a signal from the circuit itself, is required: a Keysight B2901A precision source-measure unit sources the known reference currents used in Section 5.4, and a Keysight 34470A precision digital multimeter, whose built-in four-wire resistance function is used in Section 5.5, provides voltage readings with higher accuracy than the data acquisition system. Both instruments can be controlled directly from MATLAB, which makes it possible to automate long sequences of measurements, such as the many current-voltage points needed to build the lookup tables of Section 5.4, without manual control at each step.

5.2 Data Acquisition System

All signals are acquired with a Saleae logic analyzer, shown in Figure 5.1, configured for differential analog acquisition, on 16 channels.



Figure 5.1: The Saleae logic analyzer used for data acquisition.

Every quantity of interest, both voltages taken directly from the circuit (e.g. a phase or rectifier output terminal) and the outputs of the current sensors described in Section 5.3, is captured through its own differential channel. Acquiring differentially, rather than referencing every channel to a single common ground, is necessary because several of the measured points, most notably the two terminals of a series shunt resistor, are not referenced to the same node as the rest of the circuit and would otherwise create a short-circuit. Because all channels are sampled by the same instrument, the resulting waveforms are synchronous, which is what makes it possible to pair a phase voltage with its corresponding current and compute instantaneous power.

5.3 Current Sensing: Shunt Resistors and Instrumentation Amplifiers

The Saleae itself only measures voltage, so each current of interest must first be converted into a proportional voltage. This is done with a shunt resistor R_{shunt} inserted in series with the branch to be measured, so that the current $i(t)$ flowing through it produces a voltage drop

$$v_{\text{shunt}}(t) = R_{\text{shunt}} i(t). \quad (5.1)$$

Given the low phase currents expected from the specifications of Section 1.8, $v_{\text{shunt}}(t)$ is small and, since the shunt sits in series within the branch rather than at a fixed reference node, it is not measured with respect to a common ground on either side. A sensing circuit is therefore needed to amplify the small differential

drop across the shunt to a level suitable for acquisition.

This role is carried by an INA114AU precision instrumentation amplifier connected across each shunt resistor. Figure 5.2 shows one of the six current-sensing boards actually used, built around the INA114AU, with its input terminals on the top, its gain-setting trimmer in the middle, and its output connector on the right; the sticker on the board records the actual resistance of its own shunt, measured as described in Section 5.5.

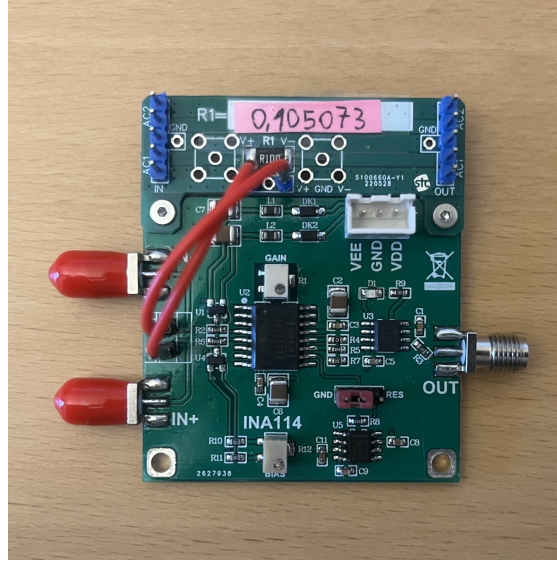


Figure 5.2: One of the six current-sensing boards used in this thesis, built around the INA114AU instrumentation amplifier.

The INA senses the two shunt terminals differentially, with high input impedance and high common-mode rejection, and delivers an output voltage proportional to the shunt voltage. Its gain G is set by a single external resistor R_G according to

$$G = 1 + \frac{50 \text{ k}\Omega}{R_G}, \quad (5.2)$$

so that the amplifier output is

$$v_{\text{out}}(t) = G v_{\text{shunt}}(t) = G R_{\text{shunt}} i(t). \quad (5.3)$$

The current sensor output voltage is then connected to the corresponding Saleae channel. In principle, the current could then be recovered by simply inverting (5.3) with the nominal values of R_{shunt} , R_G and (5.2); in practice, this is not accurate enough, and an individual calibration, described in Section 5.4, is required for every current sensor.

5.4 Current Sensors Calibration

Equation (5.3) gives the nominal, linear relationship between the sensor output and the current it measures, but a real current sensor does not follow it exactly. Component tolerances and offset voltage make every sensor, comprising its own shunt resistor and INA114AU, different from every other nominally identical sensor, and the response of the amplifier itself is not perfectly linear over the full input range. A single scale factor is therefore not accurate enough to describe a given sensor; instead, each sensor is calibrated individually by building a lookup table that maps its acquired output voltage directly to the true current.

Figure 5.3 plots the calibration curve of the six current sensors of the MP-VREH, CS1 to CS6, current driven by the B2901A against the corresponding sensor output voltage. The six sensors use nominally identical shunt resistors and INA114AU amplifiers, but their slopes are clearly different, especially for CS2. A single scale factor could not capture this spread, so each sensor needs its own lookup table.

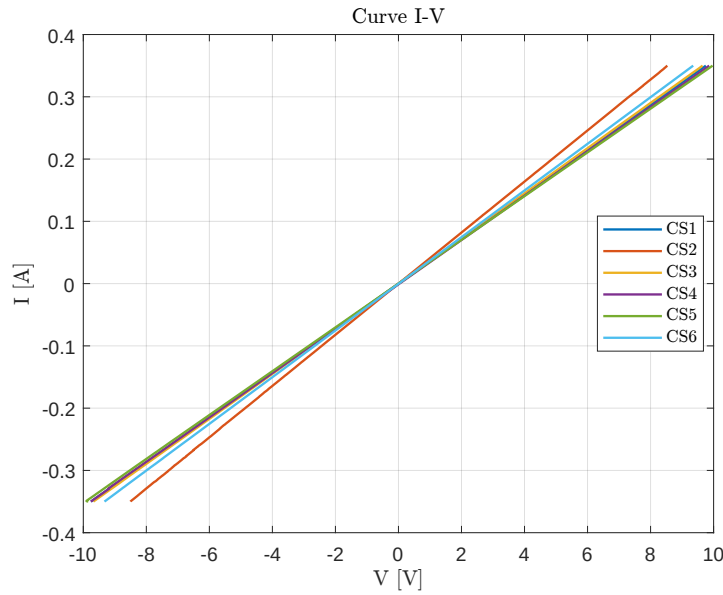


Figure 5.3: Calibration curves of the six current sensors of the MP-VREH, CS1 to CS6.

The calibration consists of driving the shunt resistor with a set of known reference currents, spanning the range expected in operation and simultaneously acquiring the corresponding sensor output voltage. Each (voltage, current) pair obtained this way becomes one entry of that sensor's lookup table. The current corresponding to any voltage acquired later during normal operation is then recovered by interpolating between the nearest entries of this table, rather than

by inverting a single linear formula, which is what allows the non-linearity and sensor-to-sensor variation described above to be compensated for. Every current used later in this thesis is recovered from the acquired voltage through this per-sensor lookup table.

The B2901A, however, has a maximum output current lower than the peak phase current found in the simulations of Chapter 4, so it cannot itself reproduce, through the $R_{\text{shunt}} = 0.1 \Omega$ shunt used on the board, the full range of shunt voltages that shunt produces in operation. The calibration shunt is therefore chosen ten times larger, $R_{\text{cal}} = 1 \Omega$. What actually sets the operating point of the INA114AU is the shunt voltage of (5.1), not the current itself. With a shunt ten times larger, the same shunt voltage, and so the same amplifier operating point, is reached with a current ten times smaller, well within the B2901A's range.

The lookup table built this way, however, relates the acquired output voltage to the current actually driven through R_{cal} during calibration, not to the current that would flow through R_{shunt} in the circuit at the same shunt voltage. At equal shunt voltage, the two currents are related by the ratio of the two resistances,

$$i_{\text{shunt}} = i_{\text{cal}} \frac{R_{\text{cal}}}{R_{\text{shunt}}}, \quad (5.4)$$

so every entry of the lookup table is rescaled by this ratio before use. Because (5.4) is a direct multiplicative correction, its accuracy depends entirely on how precisely R_{cal} and R_{shunt} are known, and their nominal values are not sufficient for this purpose. Instead, the actual resistance of every calibration and circuit shunt is measured individually, as described in Section 5.5.

5.5 Shunt Resistor Characterization

The scaling of (5.4) requires the actual resistance of the $R_{\text{cal}} = 1 \Omega$ and $R_{\text{shunt}} = 0.1 \Omega$ shunts, rather than their nominal values. At these low resistances, this cannot be obtained with an ordinary two-wire measurement, e.g. with a standard multimeter, which applies its test current and senses the resulting voltage through the same pair of cables. The resistance of these cables and the contact resistance where the probes touch the shunt terminals are then included in the measurement, and for a 0.1Ω shunt they are typically comparable in magnitude to the resistance being measured, so a two-wire reading is not accurate enough.

Each shunt is instead characterised with the 34470A's built-in four-wire, or Kelvin, resistance function: the instrument sources a known current I through the shunt via one pair of cables, and senses the resulting voltage drop through a second, separate pair of cables, connected as close as possible to the shunt's

own terminals. Because the sensing cables carry a negligible current into the multimeter’s high-impedance input, the voltage drop across their own resistance and across the contact resistance at their tips is negligible, and the voltage V it measures is essentially the drop across the shunt element alone. The 34470A then computes the shunt’s actual resistance directly from Ohm’s law,

$$R = \frac{V}{I}, \quad (5.5)$$

getting rid of the cable and contact resistance errors that limit the two-wire measurement. This four-wire measurement is applied individually to every calibration shunt and every circuit shunt used in this thesis, and the resulting actual resistances, rather than the nominal $1\ \Omega$ and $0.1\ \Omega$ values, are the ones used in the scaling of (5.4).

A single reading of (5.5) is still affected by the noise of the 34470A itself, so rather than relying on one measurement, a long sequence of repeated four-wire measurements is acquired automatically using a MATLAB script for every shunt in the system. Figure 5.4 illustrates this with one representative example, 2000 consecutive measurements of one such shunt: the top plot gives the measured resistance against acquisition number, together with the sample mean and the $\pm 1\sigma$ band, and the bottom plot shows the same 2000 values as a histogram. This scatter is simply instrument noise, which a single reading cannot cancel out. So, for every shunt, the resistance value used in the scaling of (5.4) is not any single acquisition, but the mean of a big sequence of measurements.

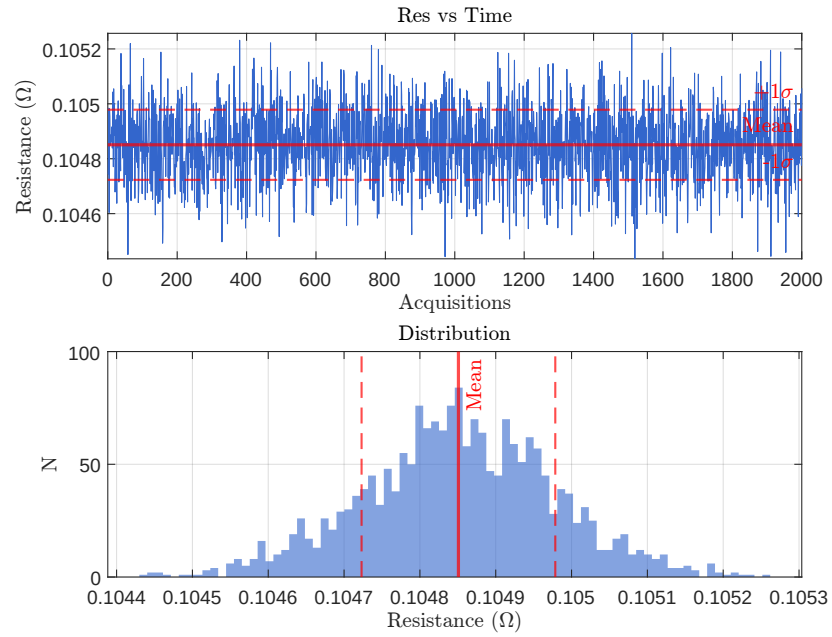


Figure 5.4: 2000 consecutive four-wire measurements of one of the shunt resistors, shown as a representative example of how the repeated measurements are distributed.

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Wish to say “Thank you”? Write it in *docs/acknowledgements.txt*.

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